Trajectory- and Population-Based Search with Path Relinking for Job-Shop Rescheduling under Sequence-Deviation Stability

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Abstract. In rescheduling, a large change from the initial schedule S_p is costly, so a revised schedule must combine efficiency with stability, that is, closeness to S_p. In this problem, S_p — stability-optimal and efficient — is in hand before the search. In a controlled comparison, we tested the hypothesis that trajectory-based search (ILS), which changes it little by little, obtains better solutions near S_p (hereafter, the high-stability region) than population-based search (Memetic), the mainstream choice. Measured by the hypervolume restricted to that region (high-stability HV), trajectory-based search exceeded population-based search in every trial of all eight scenarios; population-based search obtained almost no non-dominated solution there in three scenarios and, even in the other five, trailed by a factor of 2.0–4.5. We then equipped both structures with path-relinking-based stability-inducing operators that pull solutions back toward S_p; their effect was asymmetric. The high-stability HV of population-based search rose in every trial, matching trajectory-based search in seven of the eight, while for trajectory-based search, which obtains such solutions without the operators, the gain was limited. The best method differs by metric: population-based search with the operators for the HV of the whole front, trajectory-based search for how quickly HV rises.

Keywords: Rescheduling; Job-shop scheduling; Stability; Iterated local search; Path relinking

1. INTRODUCTION

On manufacturing shop floors, disruptions such as operation delays and machine breakdowns occur frequently and make the initial schedule difficult to execute. Responses fall into two broad classes: static approaches, which build in slack beforehand to raise the schedule's tolerance to disruptions, and dynamic approaches, which revise the schedule after a disruption occurs (Vieira et al., 2003). This study addresses the latter, specifically predictive-reactive rescheduling (Ouelhadj and Petrovic, 2009), which revises the schedule under execution so that it is feasible again.

In rescheduling, the efficiency of the revised schedule trades off against its *stability*, the degree to which it stays close to the initial schedule S_p from before the disruption. A large change incurs costs (hereafter, *schedule-change costs*): confusion on the shop floor, setup changeovers, rearrangement of materials and fixtures, reassignment of workers, and schedule changes that ripple through to downstream processes and subcontractors. Stability is therefore an objective on a par with efficiency (Wu et al., 1993; Rangsaritratsamee et al., 2004), and this study formulates the problem as multi-objective optimization of efficiency and stability. Efficiency is measured by the makespan (MS) and stability by the *sequence deviation D* from S_p (smaller is more stable; defined in Section 3.1).

Rescheduling has a peculiarity that static scheduling lacks: the solution that absorbs the delay while keeping the sequences of the initial schedule S_p is optimal for the stability objective, and it is given before the search even begins. This solution sits at D=0 on the plane with D on the horizontal axis and MS on the vertical (Figure 1), and because S_p was optimized before the disruption, this solution's MS is also small. Where schedule-change costs cannot be ignored, the solution sought lies near S_p, in what this paper calls the *high-stability region*, and shortens MS further. A method must therefore be evaluated not only by the hypervolume (HV; Zitzler and Thiele, 1999) of the whole front but also by the HV restricted to the high-stability region (hereafter, *high-stability HV*).

The job-shop scheduling problem (JSSP) addressed in this study is NP-hard, and at practical sizes metaheuristics are used rather than exact methods (Ouelhadj and Petrovic, 2009). Metaheuristics fall into two families (Blum and Roli, 2003): *trajectory-based search*, which advances by transforming a single solution little by little, and *population-based search*, which keeps several solutions and advances by recombining them. For multi-objective JSSPs, rescheduling included, population-based search built on genetic algorithms (GAs) predominates (Section 2). In this problem, however, where a high-quality initial solution S_p is given, might trajectory-based search, which changes that solution little by little, be better suited to obtaining non-dominated solutions in the high-stability region? This question is the starting point of this study.

With the local search held identical, we test this in a con-

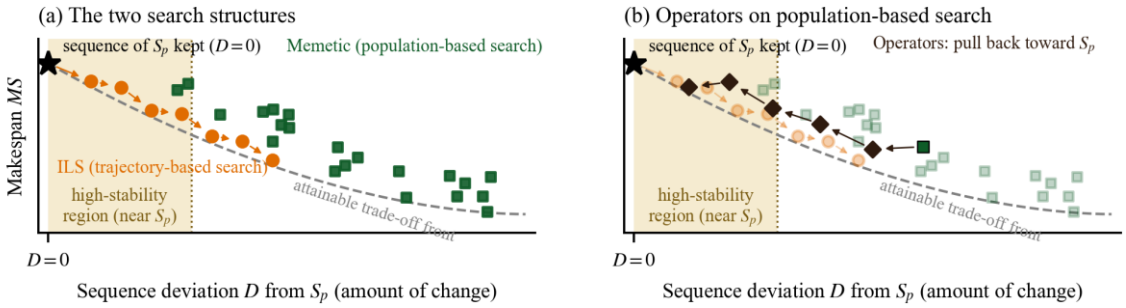


Figure 1: Schematic of the conclusions of this study.

trolled comparison between iterated local search (ILS), representing trajectory-based search, and a memetic algorithm (hereafter Memetic), a GA with local search built in, representing population-based search. We then equip both methods with *stability-inducing operators* (hereafter, operators) that pull solutions back toward S_p — that is, reduce the sequence deviation D — and examine how their effect changes with the search structure. The main findings are as follows.

• **High-stability HV is larger for trajectory-based search (ILS) than for population-based search (Memetic).** The two structures show no overlap in any of the eight scenarios: in three scenarios population-based search obtains no non-dominated solution in the high-stability region in 9–10 of the 10 trials, and in the remaining five ILS is ahead by a factor of 2.0–4.5.

• **The operators raise the high-stability HV of population-based search to match trajectory-based search, and barely raise that of trajectory-based search.** Population-based search improves in every trial of all eight scenarios, whereas trajectory-based search improves in only two of the eight, and in those two the median rises by only about 6%.

• **The best method differs by metric.** Operator-equipped Memetic is best in the HV of the whole front, ILS in how quickly HV rises, and the two are level in high-stability HV.

Figure 1 shows these findings schematically.

The aim of this study is not to win a performance race against state-of-the-art methods but to isolate the relationship between search structure and operators by posing hypotheses and controlling the conditions.

2. RELATED WORK AND HYPOTHESES

Rescheduling studies divide into those that pursue efficiency alone and those that also require stability. The former are far more numerous, as has been pointed out repeatedly (Rangsaritratsamee et al., 2004; Sun et al., 2026); the latter, the premise of this study, divide further into two approaches by how stability is secured.

Approach that embeds stability in the objective function (objective-embedding). Attempts to optimize efficiency and stability simultaneously in rescheduling began with Wu et al. (1993), who formulated both as evaluation criteria on a single machine. For the JSSP, Rangsaritratsamee et al. (2004) proposed a memetic algorithm combining a GA with local search, and Zhang et al. (2013) a hybrid of a GA and tabu search; for the flexible JSSP (FJSP), Fattahi and Fallahi (2010) likewise applied a memetic algorithm. These three combine efficiency and stability into a single scalar objective through a weight. Under Pareto dominance, Valledor et al. (2022) solve a permutation flow shop with three objectives (makespan, total weighted tardiness, and start-time deviation) by an NSGA-II variant, and Luo et al. (2025) solve a dynamic FJSP with two objectives, efficiency and stability, by a parallel NSGA-II. All treat stability as one term of the objective function.

Approach that secures stability by restricting the search space (scope-limiting). By contrast, Bean et al. (1991) proposed match-up scheduling, which reschedules only up to the point of returning to the initial schedule, and Abumaizar and Svestka (1997) proposed affected-operations rescheduling (AOR), which reschedules only the affected operations. Zakaria and Petrovic (2012) restrict the chromosome to an interval, and Sun et al. (2026) organize the range of operations allowed to be re-optimized into four levels. All have in common that stability is secured by restricting the search space.

Remaining issues. To the best of our knowledge, three issues remain across the two approaches above. First, the difference between search structures has not been examined. Across both approaches, the metaheuristics used are concentrated in population-based search (Rangsaritratsamee et al., 2004; Fattahi and Fallahi, 2010; Zakaria and Petrovic, 2012; Zhang et al., 2013; Valledor et al., 2022; Luo et al., 2025; Sun et al., 2026). No study has examined how the two search structures (trajectory-based versus population-based) differ under the rescheduling-specific property that a solution preserving the sequences of the initial schedule S_p, optimal in stability and small in MS, is in hand before the search begins. Second, no evaluation protocol reveals that difference. Comparisons have taken the form of scalar values at a few weight settings (Rangsaritratsamee et al., 2004; Zhang et al., 2013), an efficiency measure reported alongside stability (Sun et al., 2026), or indicators over the whole Pareto front such as hypervolume (Valledor et al., 2022; Luo et al., 2025); none measures the HV restricted to the high-stability region. Third, there is no move

that pulls a solution back toward S_p. The objective-embedding approach weaves stability into the *evaluation* of a solution, and the scope-limiting approach cuts the search *space* to guarantee it structurally (at the price of excluding from the search the efficiency–stability trade-off solutions that lie outside the restriction). Neither, however, has that *move* itself.

Two hypotheses. To address these issues, this study takes the high-stability HV introduced in Section 1 (the HV restricted to the high-stability region; Section 3.4) as the response and poses the following two hypotheses.

Hypothesis 1 (Figure 1(a)): trajectory-based search (ILS) starts from S_p and accumulates small changes, so it generates many solutions in the high-stability region. Population-based search (Memetic), by contrast, creates offspring by recombining the chromosomes of two parents; because recombination involves jumps that cannot be expressed as small changes from S_p, it is largely unable to generate solutions in the high-stability region. Hence, even with the same local search, *trajectory-based search attains a larger high-stability HV than population-based search*.

Hypothesis 2 (Figure 1(b)): if Hypothesis 1 holds, population-based search is failing to obtain solutions in the high-stability region. As the means of obtaining them, this study designs the move that pulls a solution back toward S_p as a stability-inducing operator (Section 3.3). On the path that pulls individuals far from S_p back to it, solutions with small sequence deviation and only a small loss in MS appear. Hence, *equipping population-based search with a stability-inducing operator raises its high-stability HV*.

Note that trajectory-based search already obtains solutions in the high-stability region without the operators, so we pose no hypothesis on it and instead equip both search structures with the same operators and compare the effect (Section 4.2).

3. PROBLEM FORMULATION AND EXPERIMENTAL DESIGN

This section casts the hypotheses of Section 2 into testable form through the problem formulation (Section 3.1) and the design of the two factors and the responses (Sections 3.2–3.4).

3.1 Problem Definition

An operation delay leaves the operation set unchanged and can be handled by readjusting the processing sequences alone, so it is the disruption under which the premise of keeping the revised schedule in the high-stability region holds most naturally. This study therefore restricts the disruption to a single operation delay (of length Δ). After such a delay occurs in an n-job, m-machine JSSP, we seek a revised schedule S_q for the initial schedule S_p. From the occurrence of the delay until the rescheduling time t_r (the time at which the delay is resolved), the shop floor continues execution by absorbing the delay without changing the machine sequences of S_p (right-shift). Operations that have started in this execution before t_r can no longer be changed and are therefore frozen. The operations from t_r onward are the optimization targets, and the decision variables are only their per-machine processing sequences.

Stability measure. Ways of measuring the change fall into two families: (i) start-time deviation and (ii) sequence deviation. Sun et al. (2026), for example, distinguish the two explicitly and adopt the former (the sum of absolute start-time deviations) as their stability measure. This study adopts the latter, because the moves used in the search (the N5 neighborhood and the swap moves of the stability-inducing operators, Sections 3.2–3.3) are all permutation moves and because sequence deviation is largely independent of MS.

$$D(S_p, S_q) = \sum_{(i,j)\in O_{opt}} \left| r_{i,j}^{p} - r_{i,j}^{q} \right| \tag{1}$$

Here $r_{i,j}$ is the processing rank of job j on machine i, and O_{opt} is the set of target operations (from t_r onward). D=0 corresponds to a solution that keeps the machine sequences of S_p.

Multi-objective formulation and scalarization. We solve $\min_{Sq}$ ($MS(S_q)$, $D(S_p,S_q)$) by a weighted sum with weight $\lambda\in[0,1]$.

$$F(S_q) = \lambda \widehat{D}(S_p, S_q) + (1-\lambda)\widehat{MS}(S_q) \tag{2}$$

Hatted symbols denote min–max normalized values, and a larger λ places more weight on stability. This study sweeps λ over several points (weighted-sum sweep) and runs an independent search at each point. The search at one point corresponds to rescheduling once in practice with the weight fixed (Rangsaritratsamee et al., 2004). Consequently, the same experiment yields both how quickly HV rises in practice, by following the search at each point over time, and the HV of the whole front, by combining the solutions of all points (Section 3.4). A weighted sum cannot reach concave parts of a non-convex front (Marler and Arora, 2010), but since all methods share the same scalarization and the solutions in concave parts visited during the search are also recorded by the UEA (Section 3.4), we consider the effect on the comparison between structures limited.

3.2 Factor 1: Search Structure — ILS and Memetic (Hypothesis 1)

To test Hypothesis 1 (Section 2), we place the contrast between ILS and Memetic at the core of the comparison, removing the confound of whether local search is present and contrasting only the search structure (trajectory-based versus population-based).

Trajectory-based search (ILS). ILS (Lourenço et al., 2019) is a framework that improves a single solution by iterating three steps: perturbing the current solution, improving the perturbed solution by local search, and deciding whether to accept the result. We adopt ILS among trajectory-based methods because local search and perturbation are separate stages. The perturbation strength, how far the search moves from the current solution at once, can be set explicitly, and the stability-

inducing operators (Section 3.3) can be placed in the perturbation stage without altering the local search or the acceptance criterion. The initial solution is S_p. The local search, shared by ILS and Memetic, is the N5 neighborhood, the standard for the JSSP (Nowicki and Smutnicki, 1996): candidates are restricted to adjacent swaps at the ends of critical blocks (moves with a prospect of improving MS), and acceptance is evaluated by $F(\lambda)$. The perturbation applies insert moves, with the strength (the number of moves) cycling in a sawtooth according to the degree of stagnation. The acceptance criterion is better acceptance: the local optimum obtained after a perturbation replaces the current solution only when it strictly improves the best $F(\lambda)$.

Population-based search (GA / Memetic). The population-based search that serves as the counterpart for Hypothesis 1 is Memetic (Neri and Cotta, 2012), a GA with local search integrated into it, which is the standard configuration in prior work. The underlying GA is a standard one in the operation-based representation (Bierwirth, 1995): a chromosome is a sequence of operations, decoded into a schedule by the GT algorithm (Giffler and Thompson, 1960) and evaluated by $F(\lambda)$. As in ILS, the initial population includes S_p as one of its individuals. In each generation, parents are chosen by tournament selection with elitism, and offspring are created by PPX crossover (Bierwirth et al., 1996) and inversion mutation. Memetic applies the N5 above to each individual of this GA and writes the improved solution back to the chromosome (Lamarckian). The plain GA without local search is a reference baseline.

3.3 Factor 2: Stability-Inducing Operators — PR and PR-*k* (Hypothesis 2)

Path relinking (PR). PR is a search method based on the idea that good intermediate solutions lie on the path connecting two high-quality solutions; it is generally combined with scatter search or the like and used to link solutions within an elite pool (Glover et al., 2000; Peng et al., 2015). We adapt it to the structure of rescheduling by *fixing the guiding solution to the single point S_p, the initial schedule*: the path runs from the current local optimum (the initiating solution) toward S_p (the guiding solution). The distance between the two is measured by the number of operations whose ranks in Eq. (1) disagree, $d=|\{(i,j)\in O_{\mathrm{opt}} : r^{p}_{i,j}\neq r^{q}_{i,j}\}|$. One step selects one disagreeing operation and swaps it with the operation currently at rank $r^{p}_{i,j}$ on the same machine, so that the selected operation comes to its rank in S_p. Repeating this reaches $D=0$ within d steps, and the operator selects intermediate solutions with small $F(\lambda)$ on this path. We fix the guiding solution to S_p because (a) S_p is the solution that minimizes the sequence deviation D, so PR has a single unambiguous interpretation as a move toward the solution minimizing D, and (b) on the path to S_p, solutions with small D and only a small loss in MS can be expected to appear, and these become the non-dominated solutions of the high-stability region. The operation to swap is chosen at random from the feasible candidates, keeping the number of evaluations at $O(d)$. Best selection, which evaluates all candidates, costs $O(d^2)$; in preliminary experiments it gained only about +1% in the HV of the whole front while taking up to about 8 times longer, so we do not adopt it.

PR-*k*. This operator is PR truncated midway. It applies the same swap move k times ($1\le k\le d$; k is the depth) to pull the solution back toward S_p, and then applies local search to the point reached, without evaluating the intermediate solutions on the path. When k equals d the solution returns all the way to S_p, and when k is small it stays near the current solution, so repeating with varying k yields solutions over a broad range between the current solution and S_p.

Adding the operators to both structures. Both structures share the core of the operators, the swap move toward S_p; when an operator fires and how its result is returned follow the standard way of incorporating operators in each structure (Table 1). In trajectory-based search (ILS), both operators act as perturbations: they fire upon stagnation, and the depth k of PR-k increases by 1 from 1 to d at each firing and resets to 1 when the upper bound is reached or the best is improved (sawtooth). In population-based search (Memetic), they are built in as an improvement step applied to each individual: after the local search, the operator is applied to each individual stochastically (the depth k of PR-k is drawn uniformly from 1 to d), and the result, re-optimized by N5, is returned to the population unconditionally; acceptance is left to tournament selection. Hereafter, the configurations without operators are called *ILS-baseline* and *Memetic-baseline*, and a configuration with an operator added is written as the baseline name plus the operator name, e.g. Memetic+PR. Each baseline together with its PR and PR-k versions is referred to as *the ILS variants* and *the Memetic variants* (three methods each).

3.4 Responses: Evaluation Metrics and Hypothesis Testing

We measure the HV of the Pareto front under an unbounded external archive (UEA; Ishibuchi et al., 2020) that keeps the non-dominated solutions among all solutions each search generates before they are subjected to selection. The recording points are aligned across the two structures: immediately after every move that changes a solution (the perturbation of trajectory-based search, the crossover and mutation of population-based search, and the stability-inducing operators of both), and after the local search. An HV over the whole front buries the difference in whether good non-dominated solutions exist in the high-stability region. This study therefore measures, in addition to this HV, the HV restricted to this region and anytime performance — the availability of a good solution whenever the search is stopped — by the area under the curve (AUC; López-Ibáñez and Stützle, 2014):

- **Overall HV**: the hypervolume over the whole non-dominated set.
- **High-stability HV**: the hypervolume restricted to the high-stability region $D<P_\theta$. P_θ is the θ-th percentile of D over

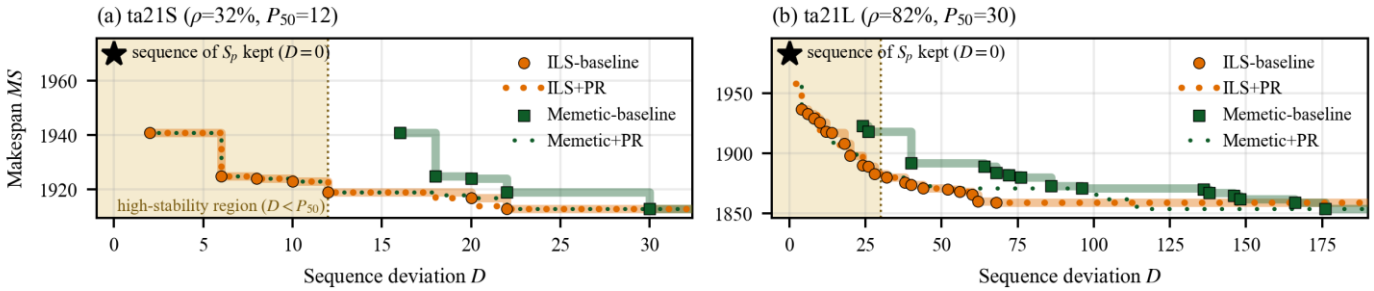


Figure 2: Actual Pareto fronts for ta21S and ta21L (non-dominated solutions of the trial with the median overall HV). PR-k is omitted because it gives almost the same front as PR in the high-stability region.

the non-dominated solutions pooled over all methods and all trials; the value of θ and its sensitivity are treated in Section 4.1 (the shaded region in Figures 1 and 2).

• **AUC**: the time average of the per-weight HV-versus-log-time curve. Because the average is taken over log time, it weights the early rise of a search heavily.

For cross-scenario comparison, HV is computed after affine normalization to $[0,1]^2$ in each scenario, with reference point (1.1,1.1). Solutions that keep the machine sequences of S_p (D=0) are trivial solutions that every method obtains without searching, so they are excluded from the HV computation. Overall HV and high-stability HV are computed after merging, in each trial, the solutions of all points of the weight sweep, whereas AUC takes the search at each weight as its unit, averaging over wall-clock log-time at each weight and then over all weights in each trial (the time window is common to all methods at each weight).

Statistics. Method comparisons within each scenario are evaluated by the Mann–Whitney U test and the effect size Cliff's δ, and all tests are two-sided. Multiple comparisons are handled by Holm correction, with the same comparison repeated across the eight scenarios forming one family; Holm-adjusted p-values are denoted p_{adj}. $|\delta|$=1.0 means no overlap: every trial of one method exceeds every trial of the other. The magnitude of a difference is reported as the ratio or relative difference of medians.

Correspondence between hypotheses and tests. Hypotheses are posed on high-stability HV alone. Hypothesis 1 is tested by comparing ILS-baseline and Memetic-baseline, which share the local search, and is regarded as supported if ILS is significantly ahead in all eight scenarios. Hypothesis 2 is tested in the same form on the gain of operator-equipped Memetic over Memetic-baseline, and is regarded as supported if the gain is significant in all eight scenarios. Trajectory-based search, on which no hypothesis is posed (Section 2), is tested in the same form: ILS with the operators against ILS-baseline.

4. COMPUTATIONAL EXPERIMENTS

This section first gives the experimental setup (Section 4.1) and then presents the results metric by metric.

4.1 Experimental Setup

We use eight scenarios obtained by imposing a completion delay of a single operation (Δ=60–148, about 0.9–1.5 times the processing time of the affected operation) on five benchmark instances (mt10, la21, la36, la40, ta21): la36S/la36M/la36L (27/54/73%), ta21S/ta21L (32/82%), mt10 (72%), la21 (35%), and la40 (32%). The percentage in parentheses is the *rescheduling ratio* $\rho=|O_{\mathrm{opt}}|/nm$, the fraction of all operations that become optimization targets under the disruption; the eight scenarios span 27–82%. For each instance the initial schedule S_p is generated by a 500-generation GA. The three la36 scenarios and the two ta21 scenarios vary only the magnitude of the disruption stepwise on the same instance with the same S_p, and are labeled S, M, and L in increasing order of ρ. The complete definition of each scenario, raw per-trial values, and tests are provided in the public repository (https://github.com/kitotakumi/stability_scheduling). The experimental settings are listed in Table 1.

Table 1: Experimental settings (common to all eight scenarios)

Item	Setting
Weight sweep	10 points, λ = 0, 0.1, …, 0.9 (λ=1.0 is excluded because it degenerates to the trivial solution at D=0)
Trials	10 per (scenario × method)
Computational budget	3000 iterations for every ILS variant, 500 generations for every GA/Memetic variant
Search	GA/Memetic variants: cx_{pb}=0.85, mut_{pb}=0.1, pop=50 (S_p plus 49 randomly generated individuals). ILS variants: insert perturbation of 2–5 moves, cycled in a sawtooth
Operators	Memetic variants: fire with probability 0.3 per individual, and PR applies local search to the top 3 intermediate solutions on the path and returns the best. ILS variants: fire first after 400 iterations without improvement and then every 10, and PR returns the single best solution on the path
Environment	AMD Ryzen 5 7530U; Python 3.12 (NumPy, DEAP, SciPy). All random seeds fixed

Validity checks. The boundary of the high-stability region is set at θ=50, so P_{50} is the median of D over the non-dominated solutions pooled across all methods and trials. The boundary is defined from the pool of compared methods, but varying θ over 33–67 leaves the conclusions of Hypotheses 1

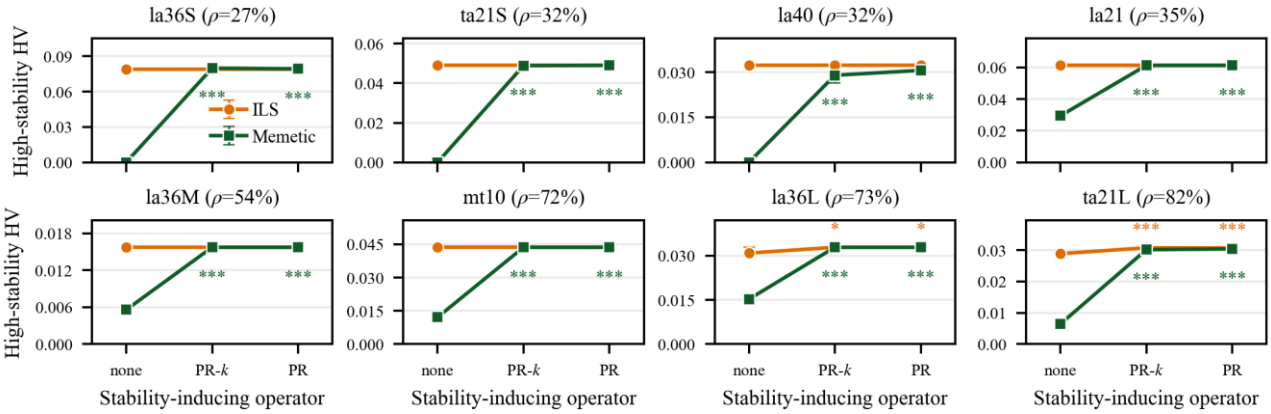


Figure 3: Search structure × stability-inducing operator (high-stability HV; medians and interquartile ranges; n=10). Symbols give significance against the configuration without operators (* p<0.05, ** p<0.01, *** p<0.001; uncorrected).

and 2 unchanged (no overlap in every scenario). Although Memetic+PR takes 4.8–9.4 times as long as ILS in wall-clock time, the median run of every method reaches 99% of its final HV within 70% of its budget, so differences in overall HV are not due to computation time. The method parameters were fixed at standard values, and the 11 axes that determine behavior, including those of the base search structures, were varied one at a time above and below the central value in two scenarios; the sensitivity (+0.5–2% in overall HV) is an order of magnitude smaller than the differences among methods (+7–54%).

4.2 High-Stability HV: ILS Leads Without the Operators, Memetic Needs Them (Hypotheses 1 and 2)

We first look at the Pareto fronts of two representative scenarios, ta21S and ta21L (the same instance under disruptions of different magnitudes), to see which methods place non-dominated solutions in the high-stability region (Figure 2). (a) In ta21S, Memetic-baseline has not a single non-dominated solution in the high-stability region, and needs D=16 to match the MS that ILS attains at D=2. (b) In ta21L it does have non-dominated solutions in the region but is inferior to ILS in MS, and the two are level only at the end where MS is smallest. In both scenarios, once the operators are added, the fronts of the two structures lie on top of each other in the high-stability region.

We check whether this picture holds in all eight scenarios (Figure 3). The gap between the points without operators corresponds to Hypothesis 1, and the slope of the Memetic line to Hypothesis 2.

• **Hypothesis 1: ILS-baseline exceeds Memetic-baseline in all eight scenarios with no overlap** (p_{adj}<0.001, $|\delta|$=1.0). In three scenarios (la36S, ta21S, la40), Memetic-baseline has almost no non-dominated solution in the high-stability region (0 points in 9–10 of the 10 trials; Figure 2(a)), and only ILS obtains non-dominated solutions there. In the remaining five scenarios, ILS is ahead by a factor of 2.0–4.5. Hypothesis 1 is therefore supported.

• **Hypothesis 2: Operator-equipped Memetic exceeds Memetic-baseline in all eight scenarios with no overlap** (p_{adj}<0.001, $|\delta|$=1.0). In the three scenarios where Memetic-baseline had almost no solution in the high-stability region, solutions appear, and in the remaining five the high-stability HV rises by a factor of 2.1–4.7. Hypothesis 2 is therefore supported. Beyond the hypothesis, comparison with ILS-baseline shows that the value reached is not significantly different in six scenarios (in five of them the medians differ by at most 1%), is 4–5% higher in ta21L, and 5–10% lower in la40 (p_{adj}<0.015). That is, with the operators added, Memetic reaches a high-stability HV at least on a par with ILS-baseline in seven of the eight scenarios. A control experiment in two scenarios (mt10, la36S), matching the strength and randomizing only the direction, tests whether this gain is an effect of perturbation in general or one specific to the direction toward S_p: a random direction reproduces only about 70% of the gain, falls short of the operators in high-stability HV with no overlap, and needs 1.7–2.4 times as long to reach the same high-stability HV.

• **Adding the operators to ILS yields almost no gain.** Operator-equipped ILS exceeds its baseline in only two of the eight scenarios (la36L, ta21L). For both PR and PR-k, p=0.013 in la36L and p<0.001 in ta21L, but after Holm correction only ta21L remains (p_{adj}<0.002; la36L is 0.091). In both scenarios the gain is small: the median rises by 6.3–6.5%. Even so, in ta21L the trials with and without the operators show no overlap ($|\delta|$=1.0). In the remaining six scenarios there is no difference.

When the operators help. That the operators act asymmetrically across the structures comes less from the difference between the structures themselves than from whether, before the operators are added, the structure has already reached the upper limit of high-stability HV. In every scenario, three to five of the seven methods lie within 2% of the best value, and this band is the upper limit attained by the compared methods (Figure 3). Memetic-baseline lies outside the band in all eight scenarios, and adding the operators brings it into the band in seven. ILS-baseline is already inside the band in six scenarios without the operators, and the only gains are in the remaining two,

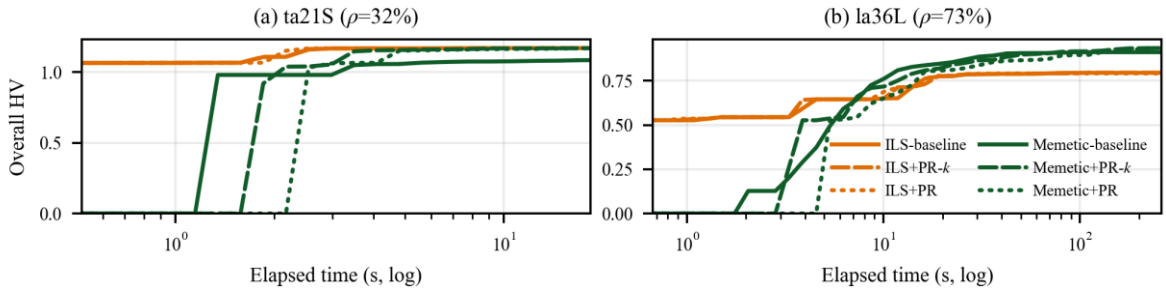


Figure 4: Anytime curves of overall HV, measured on the points visited up to time t (medians over trials). AUC is the average of the per-weight curves and does not equal the area under the weight-merged curves shown here.

which lie outside the band. Because the effect divides along the band even within the ILS variants, where the firing design is the same, the asymmetry cannot be explained by the firing design differing between the structures (Table 1).

4.3 Overall HV and AUC: The Best Method Differs by Metric

We first look at the anytime overall-HV curves of two representative scenarios, which show both the value reached and how quickly it is reached (Figure 4).

In ta21S, operator-equipped Memetic reaches the same value as ILS (1.17), but takes 6.6–7.8 s to get there against 2.5 s for ILS. In la36L, Memetic overtakes ILS at 7–10 s and reaches 0.94, whereas ILS stops at 0.80. In both scenarios, ILS rises faster, while operator-equipped Memetic reaches a final value at least as high. Table 2 compares all eight scenarios.

Table 2: Each cell gives the average rank / ARPD% (average relative percentage deviation from the best method) / number of scenarios out of 8 in which the method is level with the best, that is, its median lies within 2% of the best median. The first two are averages over the eight scenarios, which are not independent (Section 4.1), so these cross-scenario summaries are exploratory.

Method	Overall HV	High-stability HV	AUC
ILS-baseline	4.44 / 10 / 4	3.44 / 2 / 6	2.75 / 4 / 6
ILS+PR-*k*	3.56 / 9 / 4	2.62 / 0 / 8	2.62 / 5 / 6
ILS+PR	3.75 / 10 / 4	2.75 / 0 / 8	2.50 / 4 / 6
GA	6.94 / 25 / 0	6.38 / 71 / 0	7.00 / 48 / 0
Memetic-baseline	4.81 / 5 / 1	6.62 / 78 / 0	3.50 / 18 / 2
Memetic+PR-*k*	2.50 / 1 / 7	3.31 / 2 / 7	4.00 / 22 / 1
Memetic+PR	2.00 / 0 / 7	2.88 / 1 / 7	5.62 / 27 / 0

Overall HV. Operator-equipped Memetic is level with the best in seven of the eight scenarios (Table 2). As in Section 4.2, the operators raise overall HV significantly only in Memetic (in five of the eight scenarios, +3.6–11.7%), whereas for ILS the change stays within ±0.5% in every scenario. The ILS variants are level with the best in only four scenarios and fall 5–38% short in the other four. The difference in the high-stability region (Section 4.2) therefore does not reflect ILS being superior overall.

AUC. The three ILS variants are level with the best in six scenarios (Table 2). The early rise of the ILS variants can be attributed to their trajectory, which improves the single good solution S_p little by little (Figure 4). The operators do not impair this early rise in ILS, where they fire only upon stagnation: the three variants stay within 3% of one another in every scenario, with no significant difference. In Memetic, by contrast, they come at a cost: PR traces the path all the way to S_p and re-optimizes the intermediate solutions, so each call is expensive, and because it fires per individual it lowers AUC significantly in five of the eight scenarios (by 7–26%).

PR or PR-*k*. For ILS the choice hardly matters: the two differ by at most 2% on all three metrics. For Memetic, PR is slightly ahead in overall and high-stability HV (significant in two scenarios each, all favoring PR), whereas in AUC PR-*k*, whose calls are cheaper, significantly exceeds PR in six of the eight scenarios.

5. CONCLUSIONS

In rescheduling, the initial schedule S_p from before the disruption is already in hand when the search begins. With the aim of securing stability by making use of it, this study examined the relationship between search structure and stability-inducing operators through a controlled comparison over 8 scenarios × 7 methods × n=10 trials, with the local search, the objective function, and the provision of S_p all held identical. The main findings are as follows.

1. **Hypothesis 1: High-stability HV is larger for trajectory-based search (ILS) than for population-based search (Memetic).** Trajectory-based search was ahead in every trial of all eight scenarios: in three of them population-based search obtained almost no non-dominated solution in the high-stability region, and in the remaining five the gap was a factor of 2.0–4.5.

2. **Hypothesis 2 and the cross-structure comparison: The operators raise the high-stability HV of population-based search to match trajectory-based search, and barely raise that of trajectory-based search.** In high-stability HV, several methods lie within 2% of the best value, forming a band that marks the upper limit attained in this study. Without the operators, population-based search lies outside this band in all eight scenarios; with them, its high-stability HV rises in every trial and enters the band in seven. Trajectory-based search is already inside the band in six scenarios without the operators

and gains almost nothing when they are added; the only gains, of about 6% in the median, are in the remaining two, which lie outside the band. What divides the effect is thus not the difference between the structures but whether the upper limit had already been reached.

3. **Practical implication: The best method differs by metric.** Memetic with stability-inducing operators excels in overall HV, both ILS and operator-equipped Memetic in high-stability HV, and the ILS variants in AUC. Hence one chooses operator-equipped Memetic to pursue overall HV when computation time is affordable, and ILS when a revised schedule that changes little is wanted within a short computation time. Of the operators added to Memetic, PR is slightly ahead in overall HV and high-stability HV, and PR-*k* in AUC (Section 4.3).

The central contribution of this study is to establish an evaluation protocol that judges a rescheduling method by the HV restricted to the high-stability region, and to show on that basis, in a controlled comparison with the local search held identical, that population-based search, the mainstream choice, does not fully obtain the non-dominated solutions of this region and that supplying the move toward S_p as a stability-inducing operator makes up the shortfall. The effect of such an operator is decided by whether the structure it is added to has already reached the upper limit of high-stability HV. A report that an operator is effective cannot be generalized without reference to the search structure it was added to.

Limitations. (i) Comparisons whose p lies at the edge of the significance level are not settled at n=10. (ii) Validity under start-time deviation has not been examined. (iii) The disruptions are limited to operation delays. (iv) The effect of how well the initial schedule S_p is optimized has not been examined. (v) Each search structure is represented by a single method. Beyond repeating the study with these conditions varied, future work includes combining the scope-limiting approach with the operators, building stability into the crossover itself, and testing Pareto-native methods such as NSGA-II.

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